

Think Locally and Act Globally: A Frequency–Spatial Fusion Network for Infrared Small Target Detection

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Abstract—Infrared small target detection (IRSTD) remains challenging due to the extremely low signal-to-noise ratio (SNR). Existing methods struggle to balance accuracy and speed, especially under limited computational resources. To address these issues, we propose the frequency–spatial contextual fusion network (FSCFNet) based on you only look once (YOLO) v10n architecture. Particularly, the novel frequency–spatial convolution (FSConv) is designed that decomposes input features via Haar wavelet transform (WT). High-frequency cues focus on local details to highlight small targets, while low-frequency cues provide global information to complement spatial features. Subsequently, the asymmetric cross-domain attention (ACA) is developed to enhance the local central feature extraction, which reflects the typical spatial Gaussian pattern of small targets. Furthermore, we introduce the customized multiscale receptive contextual block (MRCB) to capture the long-range information by leveraging diverse dilated convolutions. In addition, the Wasserstein distance loss (WDL) is utilized to improve bounding box quality. Extensive experiments on three public datasets, including IRSTD-1k, NUDT-SIRST, and NUA-SIRST, confirm the effectiveness of FSCFNet. Notably, FSCFNet surpasses the baseline by 4.7% in precision, 3.3% in recall, and 3.9% in AP@50 on IRSTD-1k, with only a 3.6% increase in parameters. FSCFNet provides a robust solution for real-time infrared surveillance systems under resource-constrained environments. More comparisons are shown.

Index Terms—Attention mechanism, frequency–spatial domain fusion, infrared small target detection (IRSTD), remote sensing, wavelet transform (WT).

I. INTRODUCTION

INFRARED small target detection (IRSTD) is widely adopted in a wide range of applications, including military

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surveillance [1], maritime emergency [2], and fire surveillance [3]. Due to the substantial demand for infrared sensors to accurately capture thermal radiation, especially under low-illumination conditions, numerous researchers have been committed to developing effective IRSTD methods.

However, atmospheric radiation and intrinsic noise often cause infrared small targets (IRSTs) to be concealed within cluttered backgrounds [4]. Such small targets typically occupy between 0.08% and 0.58% of the total image area [5]. In addition, the low resolution and inherent imaging characteristics of infrared systems lead to a lack of color information. These factors, combined with complex backgrounds, make IRSTD highly susceptible to false alarms.

As shown in Fig. 1, object detectors based on deep learning have been extensively applied to IRSTD [6] due to their strong generalization capabilities. These approaches leverage data-driven paradigms to effectively handle a variety of complex scenarios. Nevertheless, the majority of such methods overlook informative representations in the frequency domain. This oversight limits their ability to distinguish small targets from complex backgrounds. While false alarms resemble small targets in the spatial domain, they often differ in the frequency domain. Overall, the persistent challenges in IRSTD can be attributed to these critical limitations.

- 1) *Scarcity of Discriminative Spatial Features*: IRST typically appears as isolated, low-resolution, point-like signals. These unique characteristics render localization methods that depend exclusively on spatial-domain information are often insufficient.
- 2) *Prevalence of Background Clutter and Noise Artifacts*: IRST are easily obscured by various background interferences. These interferences, including sensor noise, thermal hotspots, and structural clutter, often appear as similar-looking blobs in the image, making it difficult to differentiate true targets from background based solely on spatial-domain visual features.

Leveraging frequency-domain information helps to identify IRST against complex backgrounds. As shown in Fig. 2, we demonstrate the effectiveness of the discrete Haar wavelet transform (DWT) through a visual analysis of its decomposition results. Notably, high-frequency detailed components make IRST more distinguishable, highlighting the potential of wavelet-based frequency analysis. However, the spatial domain

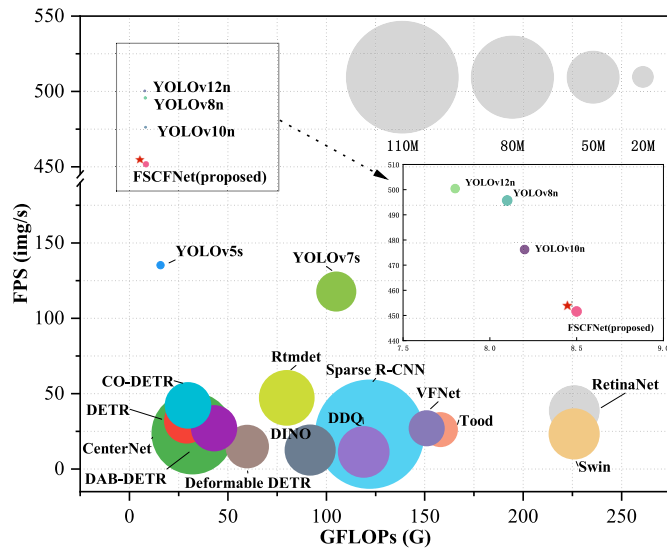


Fig. 1. Comparisons with other state-of-the-art object detectors in terms of parameters, GFLOPs, and FPS. The size of each bubble corresponds to the model parameters.

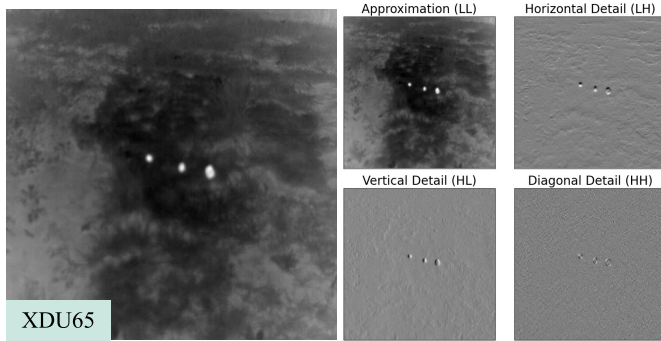


Fig. 2. (Left) Original remote sensing image from IRSTD-1k [6], with (bottom left) image index. (Right) Corresponding DWT results. The low-frequency component (LL) retains the overall structure and background information, while the high-frequency details capture directional information (horizontal, vertical, and diagonal) from the input image.

typically enables feature extraction with larger receptive fields than the frequency domain. Therefore, the two domains are inherently complementary and can work together to jointly improve IRSTD task performance.

As illustrated in Fig. 3, IRST generally present a Gaussian-like 3-D intensity distribution, characterized by a high-intensity central peak that smoothly attenuates toward the periphery. A widely adopted strategy for enhancing contextual representation is to incorporate a cross-attention mechanism, facilitating more adaptive and complementary information exchange between modalities. Transformer-based fusion frameworks have demonstrated promising performance in various vision tasks by capturing global contextual cues. However, such approaches often suffer from high computational complexity, limiting their deployment in real-time or resource-constrained scenarios. Therefore, it is essential to develop efficient alternatives that can emphasize local central features while maintaining low computational complexity.

The existing C2f module is designed to enhance feature representation. However, it exhibits limitations in effectively

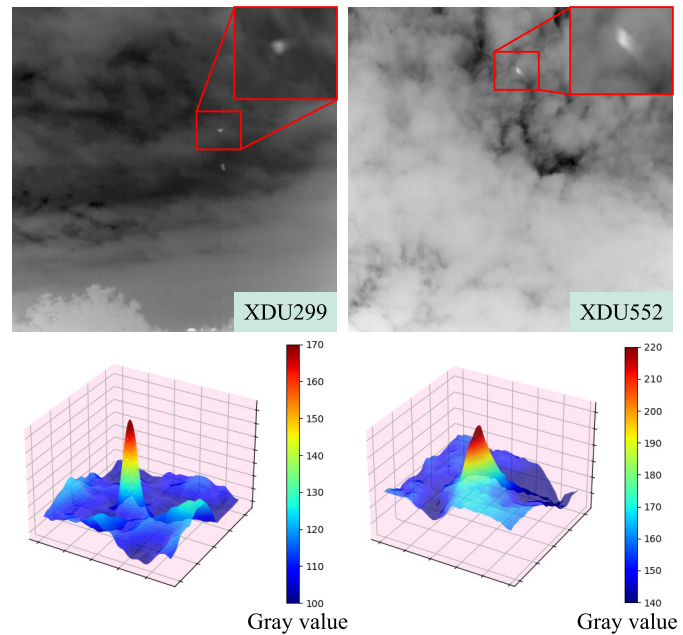


Fig. 3. Grayscale 3-D visualization of IRST from IRSTD-1k [6]. (Bottom right) Image index.

extracting features from small targets. Concretely, the C2f module fails to capture and integrate sufficient contextual information, which is critical for detecting small and low-contrast targets in infrared images. Its relatively shallow receptive field and limited multiscale perception can lead to missed or inaccurate detections of IRST.

Traditional intersection over union (IoU)-based loss functions face significant limitations in IRSTD, where annotations are discrete and pixel-level [7]. Due to the extremely small size of targets, even minor localization errors can result in zero or undefined IoU, making the loss noninformative and gradient-free. Moreover, rotated IoU is not differentiable and does not align well with IRSTD evaluation metrics, which can lead to unstable training and poor optimization results.

The Wasserstein distance loss (WDL) effectively addresses these limitations by representing each rotated bounding box as a 2-D Gaussian distribution [8], where the mean encodes the box center and the covariance matrix captures its scale and orientation. By introducing probabilistic modeling, the dissimilarity between predicted and ground-truth boxes can be measured using the normalized Wasserstein distance (NWD) in a continuous and differentiable way. As a result, WDL aligns better with detection metrics and shows improved robustness when handling small and rotation-sensitive targets commonly seen in IRSTD tasks.

Due to the extremely small size of IRST, they are often obscured by cluttered and noisy backgrounds, making accurate detection highly challenging. Moreover, existing methods that rely solely on either spatial-domain or frequency-domain features typically suffer from limited feature extraction and poor localization accuracy. To address these challenges, we propose to jointly leverage spatial-domain features and frequency-domain representations, enabling the model to effectively capture the fine-grained local structures as well as long-range

contextual dependencies for improved target discrimination. This complementary fusion enhances the localization of small targets while improving robustness against background interference. Based on these insights, as shown in Fig. 4, we design a lightweight detection framework that achieves a good balance between accuracy and computational efficiency, making it well-suited for real-time remote sensing surveillance applications.

Our contributions can be summarized as follows.

- 1) The novel plug-and-play convolution module, frequency–spatial convolution (FSConv), is designed, which integrates a DWT decomposer to capture both local details and global structural information in the additional frequency domain, preserving spatial characteristics at the same time.
- 2) The new cross-attention-based mechanism, asymmetric cross-domain attention (ACA), is proposed to facilitate the feature fusion by focusing on the local central regions of IRST, effectively strengthening their salient spatial characteristics.
- 3) Inspired by inception architecture, we propose the customized multiscale receptive contextual block (MRCB) to effectively capture long-range contextual dependencies using multiscale dilated convolutions.
- 4) Extensive experiments across multiple datasets validate that the frequency–spatial contextual fusion network (FSCFNet) maintains a lightweight design, while significantly improving both the accuracy and robustness of IRSTD.

II. RELATED WORKS

A. Traditional Methods

IRSTD methods can be broadly categorized into two primary approaches: traditional methods and deep learning-based methods [6], [9]. Traditional IRSTD methods mainly rely on handcrafted feature extraction and focus on utilizing the intensity-related characteristics, such as brightness, darkness, and local contrast. These methods generally are categorized into filter-based methods, HVS-based methods, and LRM-based methods. For instance, filter-based methods aim to reduce background clutter by leveraging various filtering techniques. Ren et al. [10] highlighted that background information is primarily concentrated in the low-frequency spectrum, whereas small targets are predominantly represented in the high-frequency components. HVS-based methods focus on enhancing small target perception by simulating human contrast sensitivity mechanisms. Wei et al. [11] developed a multiscale patches contrast measure (MPCM), which enhances small targets by adaptively computing local contrast across multiple scales. LRM-based techniques model the infrared background as a low-rank component and small targets as sparse outliers, enabling effective separation of targets from clutter to achieve accurate detection.

Despite the initial research and application achievements, traditional methods show limitations in handling increasingly complex scene variations and diverse target morphologies. In addition, target irregularities and brightness fluctuations challenge their detection accuracy.

B. Wavelet Transform (WT) in Computer Vision

Due to its capacity for multiscale decomposition, WT plays a crucial role in signal and image processing tasks [12]. It provides a multiresolution representation by capturing both low-frequency and high-frequency components of signals or images, helping to better understand their structure and features. In deep learning, methods combining CNNs and WT have been widely used in a series of tasks, such as image denoising, super-resolution, and compression.

More related to our work, Yang et al. [13] proposed PBT, which fully leveraged frequency-domain prior information to enhance the discrimination between targets and background. Finder et al. [14] proposed WT convolution (WTConv) to increase the receptive field of convolutions effectively. Yang et al. [15] developed SFFNet to bridge the semantic gap between frequency and spatial features. These works highlight the effectiveness of applying convolutional operations independently to low- and high-frequency components, which facilitates the extraction of richer and more discriminative features. Inspired by prior research and these advantages, we aim to integrate frequency-domain information while retaining spatial structures with significant grayscale changes. This joint representation enhances the performance of IRSTD by improving both target localization and background suppression.

C. Cross-Attention Mechanism

The cross-attention mechanism has emerged as a key component in modern computer vision, particularly in scenarios that involve the integration of heterogeneous sources of information. Unlike self-attention, which focuses on capturing intramodal dependencies, cross attention facilitates the alignment and interaction of relevant features between different modalities or feature spaces. Zhu et al. [16] proposed a pairwise cross attention (PWCA) to establish the interactions between image pairs. Praveen et al. [17] designed a joint cross-attention model to extract the salient features across audio-visual modalities. Tang et al. [18] developed a dual-attention Transformer (DATFuse) to fuse infrared and low-illumination visible-light images, which achieves better performance during sampling and pooling operations.

Recently, Transformers, initially proposed for natural language processing (NLP) [19], have become the dominant architecture in computer vision with the pioneering work of Vision Transformer (ViT) [20]. Subsequently, their success has been shown in various vision tasks, including image classification, object detection, and segmentation. However, the self-attention mechanism in Transformers primarily captures pairwise correlations while potentially overlooking complementary local details that are also crucial for accurate perception. To address this limitation, we integrate asymmetric convolutional kernels to generate diverse attention maps motivated by the cross-attention mechanism. This design enables CNNs to precisely focus on the central regions of IRST, thereby enhancing target localization and feature representation.

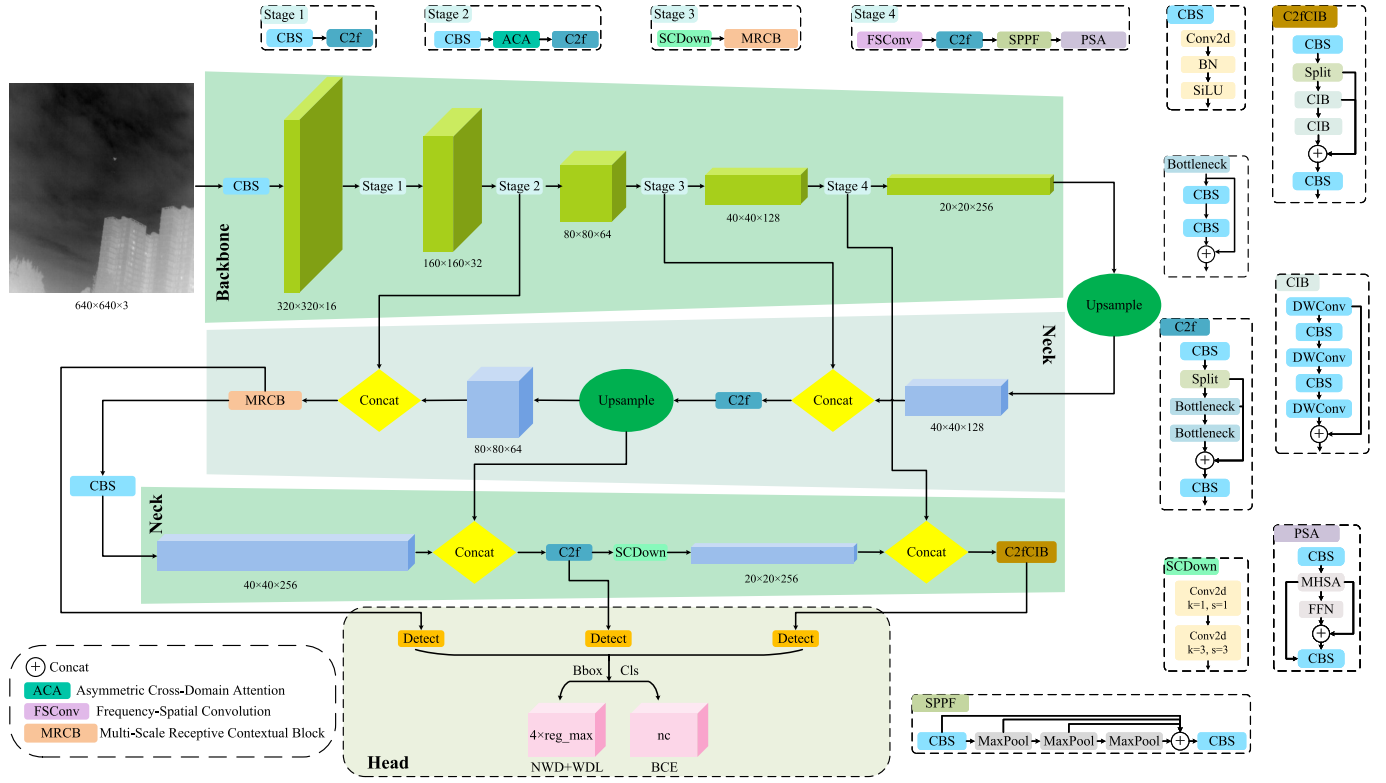


Fig. 4. Overall model architecture of FSCFNet.

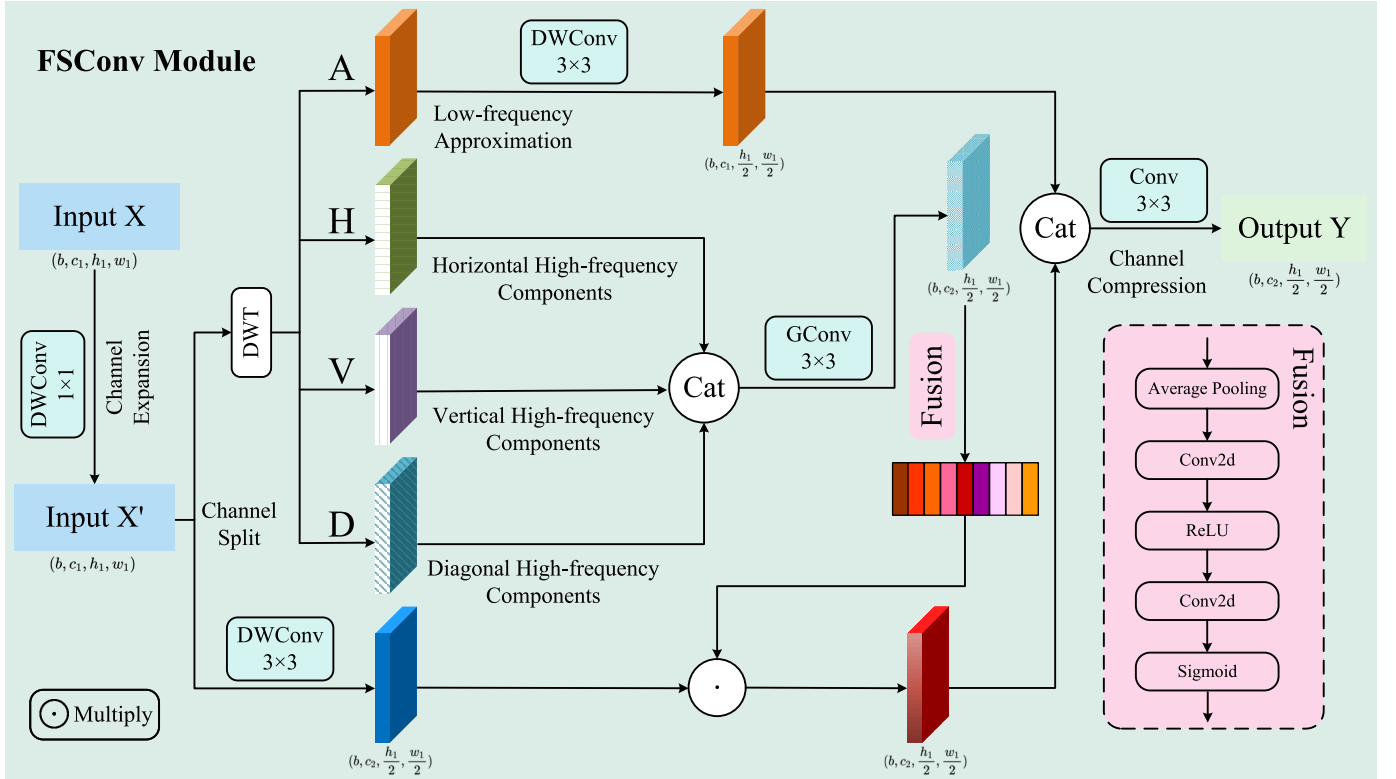


Fig. 5. Architecture of the FSCConv module.

III. METHODS

A. Frequency-Spatial Convolution

The detailed architecture of the FSCConv module can be observed in Fig. 5. To fully integrate both spatial- and

frequency-domain cues into the CNN framework, DWT is utilized to decompose the input features into approximate (low-frequency) and detailed (high-frequency) components [15]. The high-frequency components emphasize fine-grained

local details and contain key discriminative information that helps to local targets stand out from complex backgrounds [13]. This enables the network channels to more effectively focus on informative regions that are crucial for accurate detection. Meanwhile, the low-frequency component preserves the global structural information and broader contextual background, enhancing the model's robustness to large-scale clutter and spatial variations.

1) *Channel Expansion and Splitting*: Given an input tensor $X^{(b,c_1,h_1,w_1)}$, where b, c_1, h_1 , and w_1 represent the batch size, channel dimension, height, and width, respectively, a depth-wise convolution (DWConv) is utilized to double its channel dimension. This expansion generates an intermediate representation $X^{(b,2c_1,h_1,w_1)}$, effectively providing sufficient dimensional capacity for subsequent analysis of the frequency domain and the spatial domain.

2) *Frequency Branch*: Subsequently, the proposed module applies DWT to convert spatial-domain features into four frequency subbands: one low-frequency component and three high-frequency components capturing horizontal, vertical, and diagonal details

$$F_h = \text{Concat}(H, V, D) \quad (1)$$

$$F_l = \text{SiLU}(\text{BN}(A \otimes \delta^{(3,3,c_1)})) \quad (2)$$

$$F'_h = \text{SiLU}(\text{BN}(F_h \otimes \delta^{(3,3,c_2)})) \quad (3)$$

where $\otimes$ is the convolution operator and $\delta^{(k,k,c)}$ denotes the convolution with a kernel size of $k \times k$ and an output channel of c . A represents the low-frequency component and H, V , and D represent the horizontal, vertical, and diagonal high-frequency components, respectively. The three high-frequency components are then concatenated and processed using group convolution to extract the final high-frequency features. Concurrently, the low-frequency component undergoes independent processing via DWConv.

3) *Spatial Branch*: In parallel to the frequency branch, the spatial branch performs a DWConv with a kernel size 3×3 , directly extracting rich spatial information. To dynamically recalibrate channelwise feature importance, we integrate high-frequency information into a channel descriptor by utilizing a sigmoid activation function to generate channelwise attention weights

$$F_s = \text{SiLU}(\text{BN}(X \otimes \delta^{(3,3,c_2)})) \quad (4)$$

$$z_c = F_{\text{sq}}(F'_{h,c}) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F'_{h,c}(i, j) \quad (5)$$

$$F'_s = F_{\text{ex}}(z, W) = \sigma(W_2 \xi(W_1 z)) \quad (6)$$

where z denotes the channelwise global average pooling, W_1 and W_2 are the learned parameters, $\xi(\cdot)$ denotes the ReLU activation, and $\sigma(\cdot)$ is the sigmoid activation.

4) *Feature Fusion*: Finally, spatial features and low-frequency features are fused via elementwise addition, implemented using a 1×1 convolutional kernel. This approach reduces computational complexity while enhancing structural consistency

$$X' = \text{Concat}(F'_s, F_l) \quad (7)$$

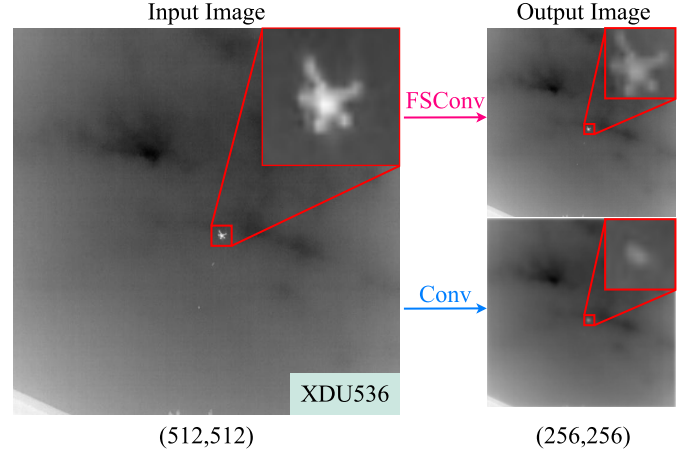


Fig. 6. Output visualization of FSCnv and standard Conv on an infrared image from IRSTD-1k [6] [(bottom right) index], showing that FSCnv has a better performance on maintaining the UAV's texture and structural details than Conv.

$$Y = \text{SiLU}(\text{BN}(X' \otimes \delta^{(1,1,c_2)})) \quad (8)$$

Notably, FSCnv utilizes depthwise and group convolutions, significantly reducing the number of parameters while effectively integrating frequency and spatial-domain information. The parameter count of a standard convolution layer can be computed using the following formula:

$$\text{Conv}_{\text{params}} = c_2 \times c_1 \times k^2, \quad (\text{bias} = \text{False}) \quad (9)$$

where k represents the convolutional kernel size. If the number of output channels (c_2) equals the input channels (c_1), the parameters of Conv are $9c_1^2$, and parameters of our FSCnv($k = 3$) are calculated as follows:

$$\text{FSCnv}_{\text{params}} = 2c_1 + 45c_2 + \frac{17}{8}c_2^2 \approx 47c_1 + 2.125c_1^2.$$

When comparing the two parameter expressions, we observe that when $c_1 < 7$, the expression $47c_1 + 2.125c_1^2$ is larger than $9c_1^2$. However, as c_1 increases beyond approximately 7, the quadratic term becomes dominant, and $9c_1^2$ surpasses the former. This indicates that the efficiency advantage of the parameters depends on the dimension size of the channel. In practical scenarios, FSCnv is typically employed to extract mid-level or high-level features in neural networks. For example, in architectures like the you only look once (YOLO) series, c_1 is usually a multiple of 64. Under this situation, FSCnv reduces the number of parameters by 68.2%, compared with a standard 3×3 Conv.

In addition, we calculate the average value across multiple channels from the outputs of the FSCnv and Conv to generate the visualizations shown in Fig. 6. The results show that FSCnv retains more IRSTD-related features during the downsampling process.

B. Asymmetric Cross-Domain Attention

The proposed ACA mechanism follows a structural design similar to the self-attention commonly used in recent Transformer-based models [21], [22]. As illustrated in Fig. 7, our main objective is to enhance the representation of IRSTD

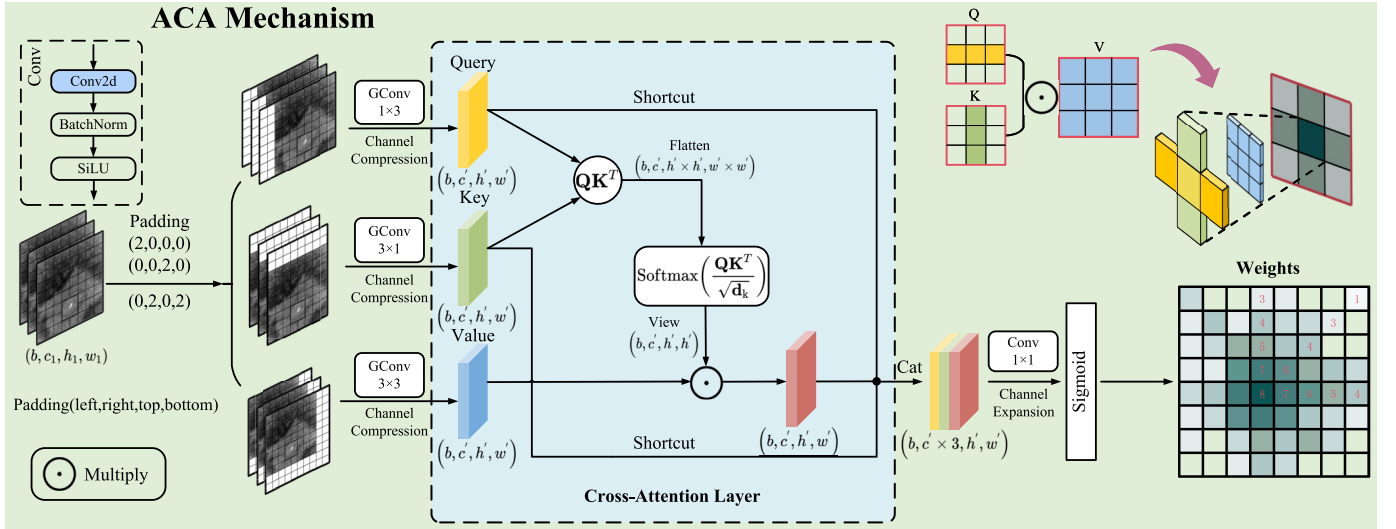


Fig. 7. Architecture of the ACA mechanism.

by reinforcing center-focused feature responses. To this end, ACA generates query, key, and value embeddings through convolutional operations with different kernel sizes, specifically using 1×3 , 3×1 , and 3×3 convolutions. This asymmetric structure of the attention mechanism is vital for accurately distinguishing IRST from complex and cluttered backgrounds.

1) *Asymmetric Padding and Feature Projection*: The initial steps involve padding the input feature map $X^{(b,c_1,h_1,w_1)}$ asymmetrically to align with the respective convolution kernel shapes

$$X_1^{(b,c',h',w')} = X_{P(2,0,0,0)}^{(b,c_1,h_1,w_1)} \otimes \delta_1^{(1,3,c')} \quad (10)$$

$$X_2^{(b,c',h',w')} = X_{P(0,0,2,0)}^{(b,c_1,h_1,w_1)} \otimes \delta_2^{(3,1,c')} \quad (11)$$

$$X_3^{(b,c',h',w')} = X_{P(0,2,0,2)}^{(b,c_1,h_1,w_1)} \otimes \delta_3^{(3,3,c')} \quad (12)$$

where the padding parameter $P(2,0,0,0)$ indicates the number of padding pixels for the left, right, top, and bottom directions, respectively. The output feature map's height (h'), width (w'), and channel count (c') after the first layer of interleaved convolution are related to the input feature map as follows:

$$h' = \frac{h_1 - 1}{s} + 1, \quad w' = \frac{w_1 - 1}{s} + 1, \quad c' = c_1 \times \text{ratio}. \quad (13)$$

Here, the ratio refers to the degree of compression, typically ranging from 0 to 1. By appropriately compressing the channel information, the computational cost can be effectively reduced while preserving essential semantic features.

2) *Query, Key, and Value Embedding*: The padded features are processed through asymmetric convolutional layers to produce the Q , K , and V feature maps

$$Q = \text{SiLU} \left(\text{BN} \left(X_1^{(b,c',h',w')} \right) \right) \quad (14)$$

$$K = \text{SiLU} \left(\text{BN} \left(X_2^{(b,c',h',w')} \right) \right) \quad (15)$$

$$V = \text{SiLU} \left(\text{BN} \left(X_3^{(b,c',h',w')} \right) \right). \quad (16)$$

Channelwise attention is computed by transposing the tensor K and performing a scaled dot-product with the tensor Q ,

which yields the attention weight tensor V' along the channel dimension

$$V' = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (17)$$

where $(1/(d_k)^{1/2})$ is the scaling factor.

Drawing inspiration from the residual network (ResNet), we adopt a structure that promotes efficient gradient flow, thus addressing the problem of vanishing gradients. Consequently, the tensors Q , K , and V' are concatenated along the channel axis

$$Y^{(b,c_2,h',w')} = \text{Concat}(Q, K, V'), \quad c_2 = c' \times 3. \quad (18)$$

3) *Feature Recalibration and Output Weighting*: The concatenated tensor is then normalized using a 1×1 convolution with an output channel dimension of c_1 . Finally, the sigmoid function serves as an attention mask to reweight the values accordingly

$$Y_1^{(b,c_1,h',w')} = \text{SiLU} \left(\text{BN} \left(Y^{(b,c_2,h',w')} \otimes \delta_1^{(1,1,c_1)} \right) \right) \quad (19)$$

$$\text{Weight} = \text{Sigmoid} \left(Y_1^{(b,c_1,h',w')} \right). \quad (20)$$

Compared with the former related works, the proposed ACA mechanism offers several key advantages. Unlike squeeze-and-excitation (SE) [23] and convolutional block attention module (CBAM) [24], our approach employs directionally diverse kernels to better extract anisotropic spatial features, which is particularly beneficial for IRST localization. Furthermore, unlike Transformer-based self-attention [25], which excels at global context modeling but often suffers from high computational complexity and diluted focus on small-scale patterns, our method offers a lightweight but effective solution that enhances center-focused feature responses while maintaining computational efficiency. These advantages make the ACA mechanism well-suited for detecting tiny infrared targets with limited spatial presence. Fig. 8 illustrates that ACA enhances the visibility of IRST by assigning higher attention weights to these targets, while simultaneously suppressing irrelevant background regions with lower weights.

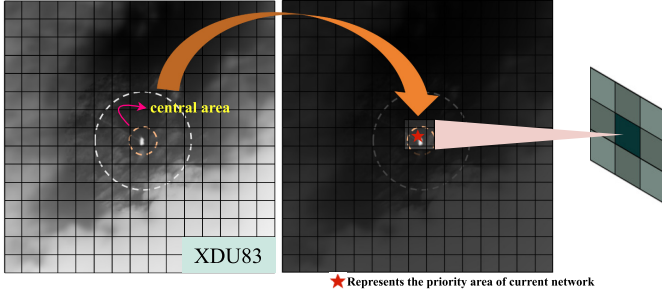


Fig. 8. Visualization of the proposed ACA reveals a strong correspondence with the pixel-level Gaussian-like spatial distribution of IRSTD from IRSTD-1K [6], with (bottom-right corner) image indices.

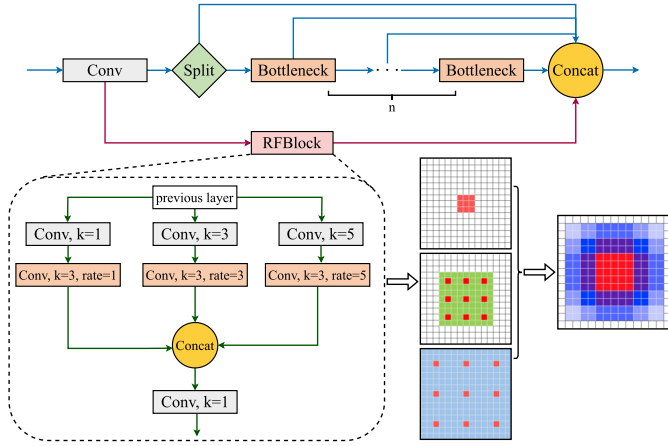


Fig. 9. Architecture of the proposed MRCB. It greatly enhances multiscale feature extraction of central information by integrating diverse receptive fields.

C. Multiscale Receptive Contextual Block

To effectively capture IRSTD embedded in cluttered backgrounds, the MRCB is designed by integrating the structural advantages of the C2f module and the receptive field block net (RFBNet) [26]. As illustrated in Fig. 9, the MRCB expands the receptive field via multibranch convolutions with diverse kernel sizes and dilation rates, thereby enhancing the network's ability to perceive contextual information at multiple scales while maintaining robustness against background noise. Specifically, we start by applying a standard convolutional layer to the input feature maps to generate an initial representation. This representation is then split into multiple parallel streams, each of which is processed by a sequence of lightweight bottlenecks. These bottlenecks retain high-level semantic information in a compact form and are applied iteratively over n stages to progressively enhance feature abstraction.

To further suppress noise potentially introduced by large dilation rates, the outputs of all branches are compressed using a 1×1 convolutional fusion layer. This layer acts as a channelwise selector, filtering out redundant or irrelevant features. In addition, a residual connection from the input is added to the fused output, scaled by a learnable factor (scale = 0.1). This stabilizes training and ensures that the contribution of dilated features remains moderate and adaptive.

The final output is then concatenated with features from earlier bottleneck paths, integrating both low-level and high-level representations. Given an input feature map $X \in \mathbb{R}^{H \times W \times C}$, the overall process of the RFBBlock can be formulated as follows:

$$X_1 = \text{Concat} \begin{pmatrix} \text{Conv}_{3 \times 3}^{r=1} (\text{Conv}_{1 \times 1}^{r=1} (X)) \\ \text{Conv}_{3 \times 3}^{r=3} (\text{Conv}_{3 \times 3}^{r=1} (X)) \\ \text{Conv}_{3 \times 3}^{r=5} (\text{Conv}_{5 \times 5}^{r=1} (X)) \end{pmatrix} \quad (21)$$

$$Y = \text{Conv}_{1 \times 1}^{r=1} (X_1) \quad (22)$$

where $\text{Conv}_{k \times k}^{r=d}(\cdot)$ denotes a convolution with kernel size $k \times k$ and dilation rate d .

Finally, the output feature map is passed downstream for the detection head. This modification leads to more robust IRSTD detection by mitigating the loss of fine-grained details and emphasizing target-relevant regions in cluttered infrared scenes.

D. Loss Function

To address the limitations of IoU-based metrics in tiny object detection, where objects occupy only a few pixels and exhibit high sensitivity to localization deviations, the WDL [27] is adopted to calculate the distance based on a new metric named NWD. By modeling bounding boxes as a 2-D Gaussian distribution and computing their similarity via the NWD, this loss provides a more robust and smooth alternative to IoU. It effectively mitigates performance degradation caused by anchor misalignment and enhances the detection accuracy for extremely small objects. Furthermore, for Gaussian distributions $\mathcal{N}_a$ and $\mathcal{N}_b$, which are modeled from bounding boxes $A = (cx_a, cy_a, w_a, h_a)$ and $B = (cx_b, cy_b, w_b, h_b)$. Let μ_a and μ_b be the spatial-statistical representations of bounding boxes A and B , respectively. Bounding boxes are encoded in vector form, specifying the center coordinates and half the size in both width and height

$$\begin{cases} \mu_a = [cx_a, cy_a, \frac{w_a}{2}, \frac{h_a}{2}]^T \\ \mu_b = [cx_b, cy_b, \frac{w_b}{2}, \frac{h_b}{2}]^T \end{cases} \quad (23)$$

then, the NWD is defined by

$$W_2^2(\mathcal{N}_a, \mathcal{N}_b) = \|\mu_a - \mu_b\|_2^2. \quad (24)$$

It is important to note that $W_2^2(\mathcal{N}_a, \mathcal{N}_b)$, representing the squared 2-Wasserstein distance between two Gaussian distributions $\mathcal{N}_a$ and $\mathcal{N}_b$, inherently serves as a distance metric. Consequently, it does not naturally conform to the characteristics of a similarity measure, which typically assumes values within a normalized range, such as $[0, 1]$, similar to the IoU. To bridge this gap and facilitate its application in similarity-based evaluations, we propose a normalized variant of the Wasserstein distance by applying an exponential transformation. This results in a bounded similarity metric, referred to as the NWD, which facilitates more interpretable and scale-consistent comparisons between distributions

$$\text{NWD}(\mathcal{N}_a, \mathcal{N}_b) = \exp \left(-\frac{\sqrt{W_2^2(\mathcal{N}_a, \mathcal{N}_b)}}{C} \right) \quad (25)$$

$$\mathcal{L}_{\text{NWD}} = 1 - \text{NWD}(\mathcal{N}_a, \mathcal{N}_b) \quad (26)$$

where C is a constant closely related to the dataset.

E. FSCFNet Structure

Fig. 4 provides a general representation of the FSCFNet framework. By jointly considering accuracy and inference efficiency, FSCFNet achieves a favorable tradeoff, demonstrating strong potential for real-worldIRSTD applications. Specifically, the backbone is divided into four stages: Stages 1–4. To enhance feature extraction in deeper layers, the original SCDOWN module in Stage 4 is replaced with the proposed FSConv module. FSConv is designed to simultaneously capture fine-grained texture details and frequency-aware spatial patterns, thereby improving the network's ability to detect small targets under complex backgrounds. This design significantly enriches the semantic representation in the deeper layers, while preserving the spatial sensitivity that is critical for detectingIRST. To better captureIRST, the ACA mechanism is integrated into the shallow stages of the backbone, enabling effective focus on salient regions while suppressing background interference. ACA dynamically enhances the response in central target areas and improves cross-scale feature focus, especially benefiting the detection of low-visibility objects. In addition, the MRCB is introduced into the mid-level layers of the backbone and deep-level layers of the neck to enable more robust contextual aggregation. By integrating multiple kernel sizes and dilation rates, MRCB captures both global and local dependencies, which is crucial for accurately distinguishingIRST from cluttered infrared backgrounds.

As illustrated in Fig. 4, the outputs of different backbone stages are fused via a PANet-style path aggregation strategy, enabling the model to integrate high-resolution spatial details from the backbone and high-level semantic abstractions from the neck. However, multiscale feature fusion often leads to feature confusion and misalignment, particularly when large-scale features dominate the representation space. To address this, we replace the original C2f module with the MRCB module in the deep-level layers of the neck.

In addition, traditional IoU-based loss functions are substituted by the WDL in the regression head, which models bounding boxes as distributions, thereby improving detection accuracy for small and ambiguous targets. By improving detection sensitivity and alignment between predictions and ground truth, FSCFNet significantly enhances the accuracy and robustness ofIRSTD, particularly under challenging remote sensing conditions.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets Description

We employed three widely recognized datasets:IRSTD-1K [6] (real dataset), NUDT-SIRST [34] (synthetic dataset), and NUAA-SIRST [9] (real dataset). These datasets serve as a rigorous and comprehensive benchmark for assessing the effectiveness and generalization capability of our method. Detailed descriptions of each dataset are as follows.

- 1) *IRSTD-1k*: TheIRSTD-1k dataset consists of 1000 manually labeled real images. The target types cover UAVs,

creatures, boats, and vehicles, and the backgrounds include the ocean, river, field, city, and sky. The rich target and background types make this dataset able to evaluate the performance of theIRSTD method in a more comprehensive way.

- 2) *NUDT-SIRST*: The NUDT-SIRST dataset contains 1327 images, including five main backgrounds: city, field, highlight, ocean, and cloud. Each image is synthesized from these real backgrounds with a variety of targets. This dataset allows for the performance of CNN-based methods to be evaluated on a wide range of target types, target sizes, and different cluttered backgrounds.
- 3) *NUAA-SIRST*: The NUAA-SIRST dataset comprises 427 infrared images. It primarily targets maritime and aerial scenarios, providing a comprehensive and challenging benchmark for evaluating the detection performance of algorithms under diverse and complex infrared conditions.

B. Implementation Details

A combination of geometric and photometric data augmentation strategies was applied to enhance the model's robustness and generalization ability. Specifically, images were randomly rotated by angles uniformly sampled from $[-10^\circ, 10^\circ]$, and fixed-angle rotations of 0° , 45° , 90° , 135° , and 180° were also applied to simulate common orientation scenarios. Translation was performed in both horizontal and vertical directions by up to $\pm 10\%$ of the image dimensions. Random scaling was applied with scale factors sampled from the range $[0.9, 1.1]$. For cropping, five independent random crops were generated per image. Each crop was centered approximately at one of five locations: the top left, top right, bottom left, bottom right, or center of the image. The crop size was randomly chosen between 80% and 100% of the original image size to simulate different object scales. To simulate illumination variations, brightness and contrast were randomly adjusted by a factor in the range $[0.8, 1.2]$ relative to the original values. In addition, Gaussian noise was added with the standard deviation σ randomly selected from the set $2.5, 5.0, 7.5, 10.0$. All augmentations were applied in a randomized sequence to maximize training sample diversity and improve robustness to real-world disturbances.

Furthermore, we divided the datasets into training and testing sets in a 4:1 ratio. During the model training, we used AdamW as the optimizer, and depending on the dataset, each dataset was trained for 500 epochs, with an initial learning rate set to 0.002, weight decay set to 0.0005, a batch size of 16, and amp set to false. All input images were resized to 640×640 . The framework was installed on a server equipped with NVIDIA GeForce RTX 3090 GPUs, which are accelerated by CUDA 11.3. We used Python 3.80 and PyTorch 1.11.0 for the software implementation.

C. Evaluation Metrics

To comprehensively evaluate the detection performance, the following metrics are adopted: Precision (P), Recall (R), F1-score (F1), average precision (AP) at both IoU thresholds

of 0.5 (AP@50) and 0.5:0.95 (AP@50:95), probability of detection (Pd), and false-alarm rate (Fa). Furthermore, we evaluate model complexity and runtime efficiency using parameter count (Params), GFLOPs, and frames/s (FPS)—critical metrics for real-time deployment. Theoretically, an effective detection method should exhibit high precision (P), recall (R), $F1$ -score ($F1$), AP, and Pd alongside low Fa and parameter count. The specific metrics are defined as follows.

1) *Precision (P)*: It can be defined as the ratio of the number of correctly predicted positive samples to the total number of samples predicted as positive by the model. The level of precision reflects the accuracy of the model in correctly identifying targets during prediction

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (27)$$

In this context, TP represents the number of true positive samples correctly identified, and FP represents the number of negative samples incorrectly identified as positive.

2) *Recall (R)*: It is defined as the model's ability to identify all targets. A high recall rate indicates that the model can detect more actual targets, resulting in fewer missed detections. Conversely, a low recall rate suggests that the model may have failed to detect many targets

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (28)$$

In this context, FN represents the number of positive samples incorrectly identified as negative.

3) *$F1$ -score ($F1$)*: It is defined as the harmonic mean of precision (P) and recall (R). A higher $F1$ -score signifies that the model not only maintains high precision but also effectively detects more targets, demonstrating better overall robustness and practicality

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (29)$$

4) *AP*: It refers to the area under the precision–recall curve, calculated at various threshold levels

$$AP = \sum_{i=1}^{n-1} (R_{i+1} - R_i) \cdot P_i, \quad P_i = \max_{j \geq i} P(j) \quad (30)$$

where R_i is the recall at the i th point, P_i is the interpolated precision corresponding to the i th recall, and n is the number of recall points. $P(j)$ represents the original precision value. This interpolation method ensures the monotonic decreasing nature of the precision–recall curve, resulting in a smooth curve, and the calculated AP better reflects the overall performance of the model.

5) *Probability of Detection (P_d)*: T_{pred} is the number of correctly predicted target instances, and T_{all} is the total number of ground-truth targets

$$P_d = \frac{T_{\text{pred}}}{T_{\text{all}}}. \quad (31)$$

6) *False-Alarm Rate (F_a)*: P_{false} denotes the number of false positive pixels, and P_{all} represents the total number of pixels

$$F_a = \frac{P_{\text{false}}}{P_{\text{all}}}. \quad (32)$$

D. Comparison With Other Convolution Module

As shown in Table I, we compare the proposed FSConv with a range of representative convolution modules. Split to be slim convolution (SPConv) [29], DWConv [28], and ghost convolution (GhostConv) [31] are primarily designed to reduce computational complexity and parameter count, making them suitable for lightweight architectures. Mixed DWConv (MixConv) [30], adaptive kernel convolution (AKConv) [32], and pinwheel-shaped convolution (PConv) [33] focus on capturing multiscale features by combining or dynamically adapting convolutional kernels. WTConv [14] expands the receptive field by incorporating WT.

In the YOLOv10n detection model, we replaced the convolution modules at layer index 7 with various alternatives for comparative experiments. As shown in Table I, most alternative convolution modules (e.g., DWConv, GhostConv, and AKConv) yielded marginal improvements and failed to consistently outperform in all evaluation metrics. Furthermore, although SPConv achieved the highest recall of 0.802 and matched the best $F1$ -score of 0.838, its precision, AP@50, and AP@50:95 were still inferior to those of our method. In contrast, the integration of the proposed FSConv module yielded the most robust and comprehensive results on the IRSTD-1K [6] dataset, achieving the highest precision of 0.897, the highest AP@50 of 0.882, and the highest AP@50:95 of 0.544 among all candidates. Similarly, in the VFNet detection model, we replaced the second layer in the bottleneck of ResNet with different convolution modules for comparison. FSConv also demonstrated superior performance, achieving the highest scores in precision of 0.732, $F1$ -score of 0.748, and AP@50 of 0.757. Notably, FSConv maintained a competitive AP@50:95 of 0.335. These results demonstrate the effectiveness of FSConv in capturing both spatial details and frequency-aware contextual information while bringing a little increase in parameters.

Fig. 10 presents the visualization results of feature representations from shallow to deep layers for different convolution types. The results demonstrate that FSConv exhibits limited effectiveness in the shallow layers, where small targets may be missed due to insufficient semantic discrimination. However, in the mid-level and deep-level stages, FSConv consistently preserves small target features while effectively suppressing irrelevant background noise, significantly outperforming conventional convolutions. By maintaining critical high-frequency details that reflect the spatial characteristics of small targets in deeper layers, FSConv reduces the risk of feature degradation or loss, thereby mitigating missed detections and enhancing localization precision.

FSConv effectively combines local detailed features and global coarse representations in the frequency domain, enabling the network to retain fine-grained structural information while enhancing contextual understanding. Specifically, the high-frequency components extracted by the DWT highlight the local variations features that are crucial for distinguishing small targets from complex infrared backgrounds. Meanwhile, the low-frequency components preserve the global structural layout and contextual semantics, enabling the network to suppress irrelevant clutter and better interpret

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT CONVOLUTION MODULES

Convolution Module	YOLOv10n detection						VFNet detection				
	P	R	F1	AP@50	AP@50:95	Params (M)	P	R	F1	AP@50	AP@50:95
Baseline	0.854	0.781	0.816	0.865	0.537	2.695	0.701	0.748	0.724	0.739	0.321
DWConv [28]	0.852	0.779	0.814	0.861	0.529	2.674	0.697	0.745	0.720	0.735	0.320
SPConv [29]	0.878	0.802	0.838	0.876	0.541	2.792	0.723	0.765	0.743	0.751	0.331
MixConv [30]	0.853	0.797	0.824	0.867	0.531	2.824	0.715	0.771	0.743	0.745	0.327
GhostConv [31]	0.859	0.783	0.819	0.866	0.534	2.823	0.714	0.757	0.735	0.742	0.325
AKConv [32]	0.862	0.776	0.817	0.859	0.527	2.765	0.725	0.743	0.734	0.735	0.319
PConv [33]	0.871	0.779	0.822	0.864	0.532	2.971	0.729	0.763	0.747	0.746	0.324
WTConv [14]	0.884	0.796	0.838	0.879	0.540	2.721	0.727	0.769	0.746	0.754	0.336
FSConv (proposed)	0.897	0.785	0.837	0.882	0.544	2.817	0.732	0.764	0.748	0.757	0.335

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT ATTENTION MECHANISMS

Attention Mechanism	YOLOv10n detection						VFNet detection				
	P	R	F1	AP@50	AP@50:95	Params (M)	P	R	F1	AP@50	AP@50:95
Baseline	0.854	0.781	0.816	0.868	0.539	2.695	0.701	0.748	0.724	0.739	0.321
SE [23]	0.879	0.765	0.818	0.871	0.543	2.695	0.726	0.749	0.737	0.746	0.322
ECA [35]	0.875	0.791	0.832	0.877	0.541	2.696	0.728	0.751	0.739	0.754	0.331
CBAM [24]	0.871	0.784	0.823	0.874	0.541	2.699	0.694	0.756	0.724	0.749	0.325
GAM [36]	0.867	0.795	0.829	0.875	0.540	2.810	0.723	0.765	0.743	0.752	0.327
EMA [37]	0.869	0.796	0.830	0.873	0.538	2.795	0.715	0.752	0.733	0.746	0.323
CAA [38]	0.872	0.801	0.836	0.876	0.543	2.879	0.721	0.768	0.744	0.756	0.334
SimAM [39]	0.861	0.793	0.826	0.869	0.539	2.695	0.695	0.752	0.723	0.741	0.321
ACA (proposed)	0.871	0.803	0.836	0.878	0.545	2.702	0.724	0.773	0.748	0.762	0.336

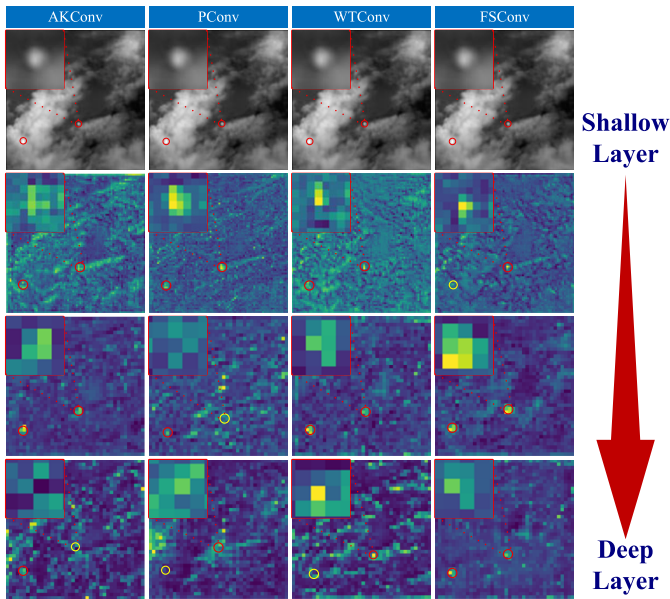


Fig. 10. Comparative feature map visualization of the proposed FSConv and other convolutions across network depths. Correct detections are marked with red circles, and missed detections are highlighted in yellow.

the overall scene. Building upon this, FSConv performs a joint frequency–spatial domain fusion strategy, where high-frequency cues are used to guide the feature responses in

the spatial domain, enhancing sensitivity to fine-scale target structures. Simultaneously, low-frequency cues act as a supplement to enhance the global understanding, improving the coherence and robustness of the feature representations. This dual-path interaction leads to more discriminative and local–global aware features, which ultimately improves IRSTD across diverse backbone networks.

E. Comparison With Other Attention Mechanism

As shown in Table II, we compare the proposed ACA with a variety of representative attention mechanisms. SE [23] and efficient channel attention (ECA) [35] focus on capturing interchannel dependencies to enhance feature representations. CBAM [24] and global attention mechanism (GAM) [36] integrate both channel and spatial attention to jointly refine feature maps across two dimensions. Efficient multiscale attention (EMA) [37] and context anchor attention (CAA) [38] are designed to capture rich contextual information by modeling long-range dependencies. In contrast, the simple, parameter-free attention module (SimAM) [39] introduces a spatial attention mechanism based on a neuroscience-inspired energy function, without introducing any additional parameters.

Notably, within the YOLOv10n detection framework, we incorporated different attention mechanisms at layer index 7 for comparative analysis. The experimental results tabulated in Table II show that most existing attention mechanisms improved specific evaluation metrics (e.g., precision or recall),

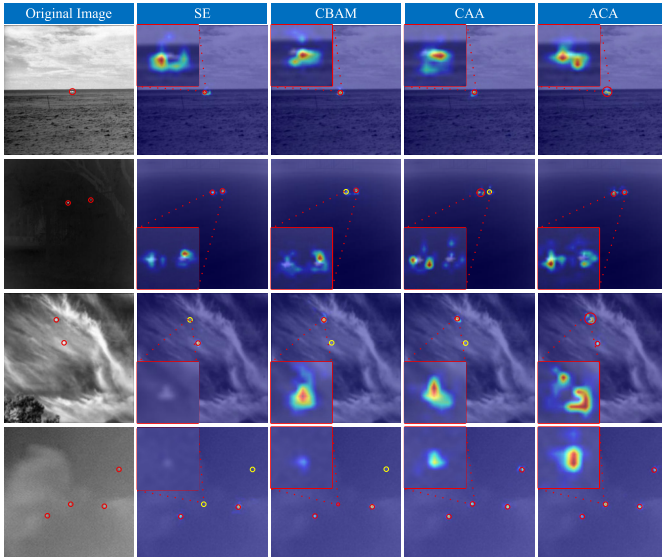


Fig. 11. Comparative CAM visualization of the proposed ACA and other attention mechanisms. Correct detections are marked with red circles, and missed detections are highlighted in yellow.

but generally did not deliver consistent gains across all key performance indicators. In contrast, the integration of ACA resulted in the most balanced and superior performance on the IRSTD-1k [6] dataset. Specifically, ACA achieved the highest recall of 0.803, tied for the best $F1$ -score of 0.836, and attained competitive values in both AP@50 of 0.878 and AP@50:95 of 0.545, all while introducing only a negligible increase in model parameters. In the VFNet detection model, ACA achieved the highest recall of 0.773, the highest $F1$ -score of 0.748, the highest AP@50 of 0.762, and the highest AP@50:95 of 0.336, while maintaining a precision of 0.724 higher than the baseline.

As illustrated in Fig. 11, the first row demonstrates that ACA generates more centered and precise attention maps, which closely align with the true target locations, thereby enhancing localization accuracy. The second, third, and fourth rows further show that compared with other attention mechanisms that often suffer from missed detections and fail to localize all targets precisely, ACA is able to consistently highlight all the small target regions with higher accuracy under cluttered or low-signal-to-noise ratio (SNR) conditions. These results indicate that ACA plays a crucial role in guiding the network's attention toward true targets, thus improving the network's performance on IRSTD under complex backgrounds.

The superior and balanced performance of ACA stems from its ability to simultaneously enhance target sensitivity and spatial precision through asymmetric cross-domain modeling. Specifically, ACA employs directional convolutions with 1×3 and 3×1 kernels to extract spatial features along the horizontal and vertical axes, capturing structural differences in different directions. In parallel, a standard 3×3 convolution captures local contextual information to provide complementary support. These heterogeneous features are fused through a cross-attention mechanism, where the query, key, and value interactions are guided not only by the strength of feature

TABLE III
COMPARISON OF DIFFERENT LOSS FUNCTIONS

Loss Function	P	R	F1	AP@50	AP@50:95
IoU [41]	0.854	0.781	0.816	0.865	0.537
DIoU [42]	0.865	0.775	0.817	0.868	0.538
CIoU [43]	0.863	0.772	0.815	0.866	0.536
GIoU [44]	0.866	0.789	0.825	0.872	0.540
WDL [27]	0.869	0.794	0.830	0.874	0.542

activations but also by the directional consistency of features across domains. To preserve spatial structure and facilitate gradient flow, ACA further applies a skip connection that adaptively combines the attention-enhanced features with the original input, ensuring that crucial details of small targets are retained.

F. Dilated Convolution Comparison in MRCB

To evaluate the effect of different dilation rate configurations within the MRCB module, we conducted a series of experiments using four representative settings: the original version (without dilation) and three multibranch configurations with dilation rates of [1, 2, 3], [3, 5, 7], and [1, 3, 5], respectively. As shown in Fig. 12, the [1, 3, 5] configuration consistently outperforms the original version across recall, precision, and mAP@50. Notably, it achieves the highest precision and mAP@50 compared with all other tested configurations. These results indicate that the [1, 3, 5] setting of dilation rates achieves a better balance between fine-grained feature extraction and contextual information aggregation.

As illustrated in Fig. 13, the overall design of MRCB is inspired by the center-periphery structure of biological receptive fields in the human visual system. Specifically, MRCB integrates multiscale features through parallel convolutional branches with different kernel sizes and dilation rates, where larger dilation rates are explicitly paired with larger kernels. Instead of relying solely on small kernels, the central region is emphasized through the combination of multiple convolutions with different dilation rates. Moreover, the final 1×1 convolution following feature concatenation serves as a learnable fusion layer that adaptively weights the contributions from each branch, effectively suppressing irrelevant or noisy activations introduced by distant context. Therefore, MRCB enables the network to focus on small targets with high attention in the center, while still leveraging long-range information to enhance global awareness.

G. Comparison With Different Loss Functions

To validate the superior performance of WDL in addressing issues caused by pixel-level discretization, especially in IRSTD, five comparative experiments were conducted. As shown in Table III, WDL achieves the best performance across all metrics, with the highest precision of 0.869, the highest recall of 0.794, the highest $F1$ -score of 0.830, the highest AP@50 of 0.874, and the highest AP@50:95 of 0.542. These results clearly demonstrate the effectiveness of WDL for IRSTD tasks, where conventional IoU-based losses struggle

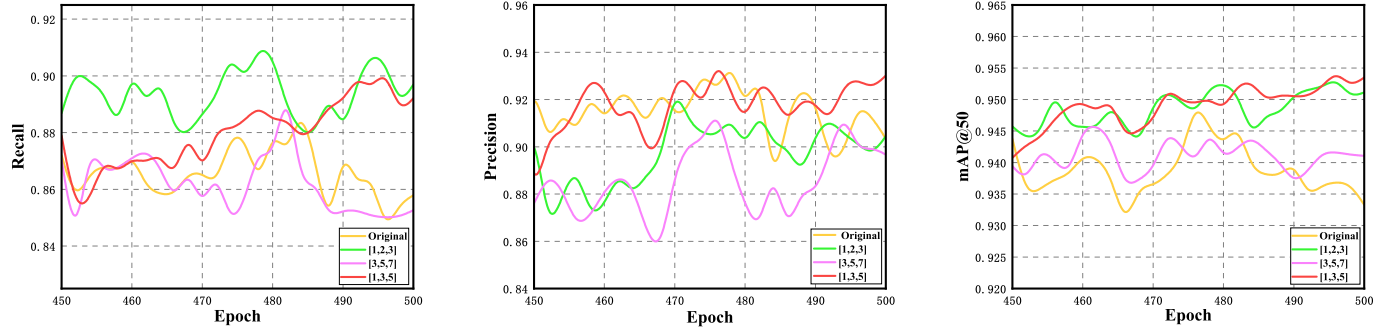


Fig. 12. Performance comparison of different dilation rate settings in MRCB.

TABLE IV

COMPARISON WITH OTHER DETECTION METHODS ON THREE PUBLIC DATASETS. RED, BLUE, AND ORANGE REPRESENT THE BEST, SECOND-BEST, AND THIRD-BEST METRICS IN EACH COLUMN, RESPECTIVELY

Year	Detection Methods	IRSTD-1K							NUDT-SIRST							NUAA-SIRST						
		P	R	F1	AP@50	AP@50:95	Pd	Fa (10^{-4})	P	R	F1	AP@50	AP@50:95	Pd	Fa (10^{-4})	P	R	F1	AP@50	AP@50:95	Pd	Fa (10^{-4})
2017	RetinaNet [45]	0.484	0.716	0.577	0.501	0.135	0.716	0.523	0.911	0.840	0.874	0.857	0.561	0.840	0.098	0.762	0.868	0.811	0.802	0.335	0.868	0.216
2019	CenterNet [46]	0.882	0.760	0.817	0.762	0.327	0.760	0.116	0.938	0.902	0.920	0.946	0.647	0.902	0.061	0.847	0.824	0.836	0.769	0.393	0.824	0.163
2020	YOLOv5s [47]	0.877	0.811	0.843	0.846	0.410	0.811	0.111	0.946	0.917	0.931	0.955	0.786	0.917	0.050	0.927	0.894	0.910	0.915	0.448	0.894	0.091
2020	Sparse R-CNN [48]	0.856	0.808	0.832	0.732	0.288	0.808	0.137	0.945	0.907	0.926	0.948	0.564	0.907	0.046	0.925	0.853	0.888	0.876	0.378	0.853	0.143
2020	DETR [49]	0.478	0.893	0.623	0.327	0.122	0.893	0.537	0.732	0.897	0.808	0.820	0.421	0.897	0.256	0.376	0.720	0.494	0.437	0.254	0.720	0.626
2021	Swin [50]	0.763	0.501	0.607	0.567	0.272	0.501	0.255	0.836	0.694	0.758	0.588	0.437	0.694	0.155	0.879	0.781	0.827	0.602	0.310	0.781	0.102
2021	Tood [51]	0.785	0.733	0.758	0.747	0.325	0.733	0.203	0.952	0.907	0.929	0.956	0.659	0.907	0.057	0.905	0.841	0.872	0.879	0.383	0.841	0.121
2021	VFNet [52]	0.688	0.768	0.726	0.740	0.321	0.768	0.304	0.946	0.906	0.926	0.944	0.660	0.906	0.051	0.836	0.814	0.825	0.835	0.391	0.814	0.179
2021	Deformable DETR [53]	0.807	0.823	0.815	0.799	0.315	0.823	0.188	0.935	0.918	0.922	0.956	0.573	0.918	0.057	0.874	0.858	0.866	0.873	0.313	0.858	0.137
2022	YOLOv7s [54]	0.796	0.749	0.772	0.762	0.318	0.749	0.189	0.924	0.878	0.901	0.917	0.736	0.878	0.072	0.891	0.846	0.868	0.913	0.374	0.846	0.115
2022	Rtmdet [55]	0.764	0.568	0.652	0.626	0.291	0.568	0.230	0.814	0.675	0.738	0.729	0.440	0.675	0.193	0.853	0.706	0.773	0.719	0.364	0.706	0.141
2022	DAB-DETR [56]	0.685	0.701	0.693	0.650	0.238	0.701	0.295	0.858	0.726	0.787	0.861	0.547	0.726	0.124	0.724	0.661	0.695	0.611	0.230	0.661	0.303
2023	DINO [57]	0.853	0.805	0.830	0.824	0.361	0.805	0.143	0.931	0.906	0.868	0.949	0.652	0.906	0.051	0.893	0.845	0.870	0.889	0.377	0.845	0.093
2023	YOLOv8n [58]	0.886	0.810	0.848	0.881	0.548	0.810	0.108	0.937	0.901	0.918	0.959	0.786	0.901	0.069	0.921	0.848	0.883	0.918	0.459	0.848	0.073
2023	DDQ [59]	0.890	0.792	0.838	0.889	0.536	0.792	0.123	0.947	0.899	0.922	0.941	0.654	0.899	0.065	0.919	0.882	0.901	0.894	0.389	0.882	0.075
2023	CO-DETR [60]	0.635	0.652	0.643	0.568	0.171	0.652	0.352	0.913	0.894	0.906	0.939	0.472	0.894	0.068	0.692	0.711	0.701	0.661	0.234	0.711	0.321
2024	YOLOv10n [61]	0.854	0.786	0.818	0.871	0.542	0.786	0.118	0.939	0.886	0.912	0.953	0.774	0.886	0.062	0.913	0.857	0.884	0.912	0.445	0.857	0.092
2025	YOLOv12n [62]	0.879	0.803	0.839	0.877	0.543	0.803	0.102	0.944	0.908	0.925	0.962	0.788	0.908	0.053	0.868	0.913	0.890	0.917	0.452	0.913	0.108
2025	FSCFNet (proposed)	0.901	0.814	0.856	0.904	0.558	0.814	0.101	0.947	0.910	0.928	0.959	0.783	0.910	0.051	0.921	0.877	0.898	0.923	0.452	0.877	0.084

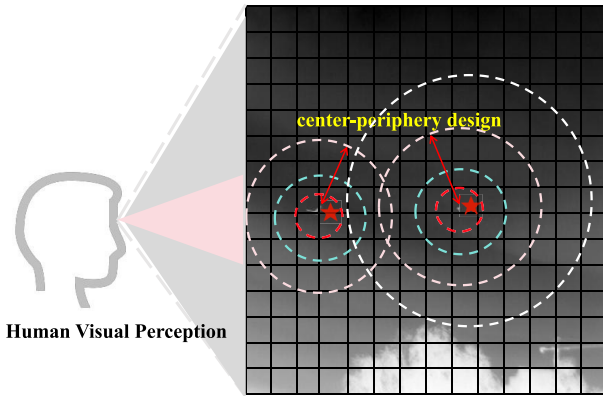


Fig. 13. Schematic of the center-aware strategy implemented by the MRCB module.

with insufficient overlap or unstable gradients. Thus, WDL is adopted as the loss function for FSCFNet to better address class imbalance and discrete target regions.

H. Comparison With Other Methods

A comprehensive evaluation of FSCFNet on three different datasets is presented in Table IV, offering insights into its generalization and adaptability. The parameters, GFLOPs, and

TABLE V

COMPARISON OF DIFFERENT METHODS IN TERMS OF PARAMETERS, GFLOPs, AND FPS

Year	Methods	Params (M)	GFLOPs (G)	FPS (frames/s)
2017	RetinaNet [45]	47.369	226	38.8
2019	CenterNet [46]	78.56	32.111	23.6
2020	YOLOv5s [47]	7.022	15.9	135.1
2020	Sparse R-CNN [48]	106	122	23.1
2020	DETR [49]	41.555	29.17	31.7
2021	Swin [50]	47.369	226	23.2
2021	Tood [51]	32.018	158	26.0
2021	VFNet [52]	32.709	151	26.9
2021	Deformable DETR [53]	40.099	59.905	14.8
2022	YOLOv7s [54]	37.197	105.1	117.6
2022	Rtmdet [55]	52.255	79.951	47.2
2022	DAB-DETR [56]	43.702	43.099	26.9
2023	DINO [57]	47.54	91.793	12.5
2023	YOLOv8n [58]	3.006	8.1	500.0
2023	DDQ [59]	48.306	119	11.2
2023	CO-DETR [60]	43.448	42.133	29.8
2024	YOLOv10n [61]	2.721	8.2	476.2
2025	YOLOv12n [62]	2.672	7.8	495.7
2025	FSCFNet (proposed)	2.820	8.5	451.6

FPS of each compared method are summarized in Table V. Among them, YOLOv12n achieves the lowest computational complexity. The proposed FSCFNet ranks third in the number of parameters and fourth in GFLOPs, outperforming most other methods in both aspects.

TABLE VI
ABLATION EXPERIMENTS FOR FSConv, ACA, MRCB, AND WDL IN FSCFNet

FSConv	ACA	MRCB	WDL	P	R	F1	AP@50	AP@50:95
				0.854	0.781	0.816	0.865	0.537
✓				0.897 (4.3% ↑)	0.785 (0.4% ↑)	0.837 (2.1% ↑)	0.882 (1.7% ↑)	0.544 (0.7% ↑)
	✓			0.871 (1.7% ↑)	0.803 (2.2% ↑)	0.836 (2.0% ↑)	0.878 (1.3% ↑)	0.545 (0.8% ↑)
		✓		0.892 (3.8% ↑)	0.778 (0.3% ↓)	0.831 (1.5% ↑)	0.875 (1.0% ↑)	0.546 (0.9% ↑)
			✓	0.859 (0.5% ↑)	0.794 (1.3% ↑)	0.825 (0.9% ↑)	0.874 (0.9% ↑)	0.542 (0.5% ↑)
✓	✓			0.874 (2.0% ↑)	0.801 (2.0% ↑)	0.836 (2.0% ↑)	0.883 (1.8% ↑)	0.544 (0.7% ↑)
✓		✓		0.895 (4.1% ↑)	0.782 (0.1% ↑)	0.835 (1.9% ↑)	0.885 (2.0% ↑)	0.541 (0.4% ↑)
	✓	✓		0.887 (3.3% ↑)	0.802 (2.1% ↑)	0.842 (2.6% ↑)	0.889 (2.4% ↑)	0.548 (1.1% ↑)
✓	✓	✓		0.893 (3.9% ↑)	0.801 (2.0% ↑)	0.845 (2.9% ↑)	0.897 (3.2% ↑)	0.553 (1.6% ↑)
✓	✓	✓	✓	0.901 (4.7% ↑)	0.814 (3.3% ↑)	0.856 (4.0% ↑)	0.904 (3.9% ↑)	0.558 (2.1% ↑)

1) *On the IRSTD-1k Dataset:* FSCFNet consistently outperforms other methods across most evaluation metrics. Although its recall (R) and Pd are slightly lower than the deformable DETR, FSCFNet exhibits superior overall performance on the other metrics. Specifically, FSCFNet achieves the highest performance in terms of precision, *F1*-score, AP@50, AP@50:95, outperforming the second-best method by 1.1%, 0.8%, 1.5%, 1.0%, respectively. Furthermore, FSCFNet achieves the lowest false-alarm rate (Fa) across all evaluated methods, confirming its robustness in distinguishing true targets from background noise.

2) *On the NUDT-SIRST Dataset:* FSCFNet shows promising performance across most evaluation metrics. It ranks second in precision (P) and AP@50, and third in recall (R), *F1*-score, AP@50:95, Pd, and Fa.

3) *On the NUAASIRST Dataset:* FSCFNet demonstrates overall superiority in this experimental set. Although FSCFNet shows relatively modest performance in recall (R) and probability of detection (Pd), it ranks third in precision (P), *F1*-score, and false-alarm rate (Fa), and achieves the highest score on the AP@50 metric. Specifically, it outperforms the second-best method by 0.5% on the AP@50 metric, highlighting its strong capability to detect small infrared targets with moderate localization accuracy requirements.

I. Ablation Study

1) *Integration of FSConv, ACA, MRCB, and WDL:* Table VI displays the ablation study results for the integration of FSConv, ACA, MRCB, and WDL into our network. The results clearly demonstrate that each individual component contributes positively to the overall performance. Notably, the full integration of all four modules exhibits the best results across all evaluation metrics. Compared with the baseline, the integrated model achieves improvements of 4.7%, 3.3%, 4.0%, 3.9%, and 2.1% in precision, recall, *F1*-score, AP@50, and AP@50:95, respectively, highlighting the effectiveness and comprehensive performance improvement of our FSCFNet. Furthermore, Table VII presents the computational complexity of each proposed module. The results show that each module introduces only a little increase in parameters and GFLOPs. Even when all modules are integrated, the total complexity increases from 2.721M to 2.820M Params and from 8.2 to

TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON OF THE PROPOSED MODULES VIA ABLATION EXPERIMENTS

Methods	Params (M)	GFLOPs (G)
YOLOv10n	2.721	8.2
YOLOv10n+FSConv	2.728	8.2
YOLOv10n+ACA	2.805	8.3
YOLOv10n+MRCB	2.729	8.3
YOLOv10n+FSConv+ACA	2.812	8.3
YOLOv10n+FSConv+ACA+MRCB	2.820	8.5

8.5 GFLOPs, demonstrating the efficiency of the proposed modules.

As shown in Table VI, the sixth row demonstrates that the ACA and FSConv module effectively enable the network to jointly leverage spatial attention and frequency enhancement. Concretely, given that IRST is easily confused with complex backgrounds or obscured by noise, ACA first guides the network to concentrate on critical local spatial regions, effectively suppressing irrelevant background interference. After ACA effectively focuses on small targets, FSConv further reinforces their high-frequency details while integrating global low-frequency information to enrich the spatial context. This helps to compensate for potential information loss during deeper semantic abstraction and enhances the model's ability to preserve fine-grained target features. Building upon the effective separation between small targets and background achieved by ACA and FSConv, MRCB further strengthens salient local features and enhances long-range contextual understanding, effectively integrating local details with global semantics. The ninth row demonstrates that ACA, FSConv, and MRCB collaboratively constitute a tightly integrated processing pipeline, yielding consistent improvements across all evaluation metrics. Finally, the WDL loss function enhances localization precision and strengthens the overall detection reliability by refining the feature representations of the first three modules, which results in a smoother bounding box regression process and thus improves the robustness of the entire detection framework.

2) *Different Strategies of Our Proposed Modules:* Table VIII illustrates that placing the FSConv module in the deep-level layers of the backbone consistently yields better overall performance, with the exception of a slight improvement in recall. Due to the low SNR between the targets and the

TABLE VIII
DIFFERENT PLACEMENT STRATEGIES FOR FSConv MODULE

layer1	layer3	layer5	layer7	P	R	F1	AP@50	AP@50:95
				0.854	0.781	0.816	0.865	0.537
✓				0.881	0.769	0.821	0.867	0.535
	✓			0.872	0.757	0.811	0.862	0.528
		✓		0.882	0.794	0.836	0.879	0.542
			✓	0.897	0.785	0.837	0.882	0.544

TABLE IX
DIFFERENT PLACEMENT STRATEGIES FOR ACA MECHANISM

layer1	layer3	layer5	layer7	P	R	F1	AP@50	AP@50:95
				0.854	0.781	0.816	0.865	0.537
✓				0.893	0.754	0.818	0.872	0.539
	✓			0.878	0.796	0.835	0.878	0.542
		✓		0.874	0.803	0.837	0.881	0.551
			✓	0.866	0.772	0.816	0.865	0.535

background in infrared images, features extracted in the shallow layers of the network often contain numerous background noise. Placing FSConv at the early stages may introduce such irrelevant high-frequency components, leading to increased false positives. At deeper layers, high-frequency components carry more abstract, semantically meaningful structural cues that are vital for localizing small and dim targets, whereas low-frequency components contribute to modeling the global context and suppressing background interference. FSConv leverages this complementary information by adaptively integrating frequency components, allowing the network to retain fine-grained localization capability while enhancing background modeling.

Table IX presents the impact of inserting the ACA module at different depths within the backbone. The results indicate that ACA is most effective when applied to mid-level layers. IRST are often indistinguishable in local patches but stand out when evaluated in a broader spatial context. Placing MRCB in deeper layers, particularly within the neck, allows the network to better differentiate small targets from complex backgrounds and extract more informative semantic cues that contribute to global contextual understanding. In contrast, shallow or mid-level layers are dominated by textures and noise, where the broad receptive fields of MRCB may inadvertently capture both targets and background clutter. By operating on high-level semantic features, MRCB leverages its global modeling capability to enhance discriminative local small targets and suppress irrelevant information, increasing localization precision and reducing false alarms.

As shown in Table X, placing the MRCB module at deeper semantic levels, particularly within the neck stage, leads to the most significant improvements in both detection accuracy and overall evaluation metrics. IRST are often indistinguishable in local patches but stand out when evaluated in a broader spatial context. Placing MRCB in deeper layers, particularly within the neck, allows the network to better differentiate small targets from complex backgrounds and extract more informative semantic cues that contribute to global contextual understanding. In contrast, shallow or mid-level layers may

TABLE X
DIFFERENT PLACEMENT STRATEGIES FOR MRCB MODULE

layer2	layer4	layer6	layer13	P	R	F1	AP@50	AP@50:95
				0.854	0.781	0.816	0.865	0.537
✓				0.897	0.770	0.828	0.876	0.535
	✓			0.872	0.775	0.823	0.866	0.534
		✓		0.888	0.786	0.836	0.879	0.545
			✓	0.875	0.798	0.836	0.882	0.552

contain a significant amount of low-level noise. Therefore, the broader receptive fields of MRCB may capture both targets and noise at the same time, reducing localization precision and increasing false detections.

J. Visualization Experiment

Fig. 14 presents the visualization results of various methods applied to several representative infrared images. These visualizations cover a wide variety of background types, such as urban, ocean, field, cloud, and highlight, showcasing the robustness of FSCFNet across different scenes. For instance, the first and second rows show that in urban and highlight backgrounds, FSCFNet effectively suppresses noise from buildings or strong intensity areas. The third and fourth rows show that in ocean and field backgrounds, it is capable of preserving fine-grained target details under cluttered backgrounds and low-contrast conditions. In cloud scenes, it minimizes false positives while maintaining precise localization. In field environments, FSCFNet demonstrates strong background suppression and maintains detection stability even under low-contrast and noisy conditions, as shown in the vehicle rows. In terms of target types, FSCFNet handles small and dim objects such as UAVs and creatures with high precision, accurately detecting relatively larger and more ambiguous objects like boats and vehicles at the same time. Traditional detectors such as RetinaNet, VFNet, and Rtmddet often produce inaccurate bounding boxes and generate false alarms in the urban, ocean, and highlight scenes, with RetinaNet and VFNet in particular exhibiting a tendency toward missed detections. Similarly, detectors from the YOLO series, including YOLOv5s, YOLOv8n, and YOLOv12n, tend to trigger false reports and struggle to achieve high detection accuracy in cloud-dominated scenes. In complex field environments, where vegetation and texture clutter introduce significant background noise, traditional detectors and YOLO series models always suffer from false alarms or fail to detect small and low-contrast targets accurately, leading to frequent missed detections and poor localization performance. In contrast, the proposed FSCFNet demonstrates more accurate and complete detection results, with predicted bounding boxes closely aligning with the ground-truth annotations. These visualizations provide additional evidence that FSCFNet enhances the contrast between small targets and cluttered backgrounds, thereby improving detection robustness in challenging infrared environments.

K. Generalization Experiment

As shown in Table XI, the proposed variants of FSCFNet consistently outperform their baseline counterparts

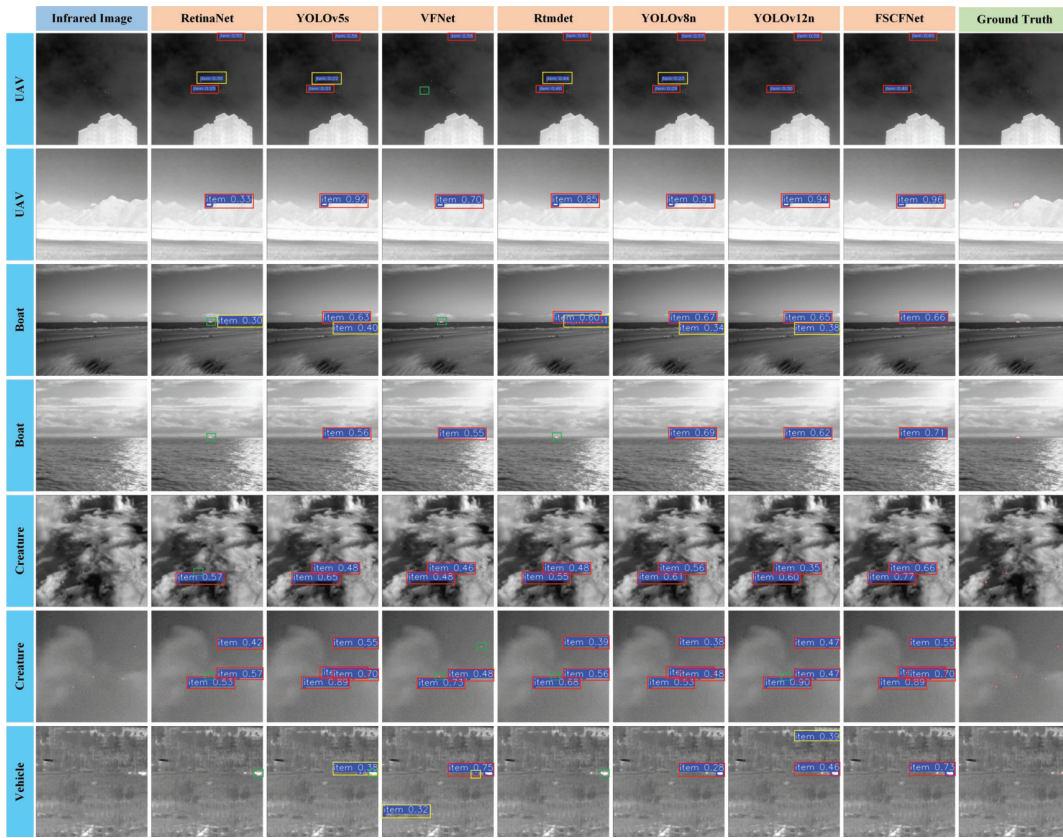


Fig. 14. Visualization comparison of detection results via different methods on representative images from IRSTD-1k [6] and NUDT-SIRST [34] datasets. The red, yellow, and green boxes denote correct detection, false alarms, and missed detections, respectively.

TABLE XI
COMPARISON OF FSCFNET WITH YOLOV10 VARIANTS

Model	P	R	F1	AP@50	AP@50:95	Params (M)
YOLOv10n	0.854	0.781	0.816	0.865	0.537	2.721
FSCFNet-n	0.901	0.814	0.856	0.904	0.558	2.820
YOLOv10s	0.874	0.790	0.830	0.876	0.572	8.036
FSCFNet-s	0.898	0.816	0.855	0.911	0.589	8.559
YOLOv10m	0.915	0.784	0.843	0.894	0.618	16.45
FSCFNet-m	0.927	0.812	0.867	0.918	0.623	17.16
YOLOv10l	0.936	0.799	0.861	0.921	0.663	25.77
FSCFNet-l	0.941	0.818	0.874	0.937	0.671	26.73

of YOLOv10 in all evaluation metrics. Specifically, FSCFNet-n achieves the most significant improvements, yielding higher AP@50 and AP@50:95 scores while maintaining a low parameter count, thus demonstrating its superior efficiency. In addition, other variants (e.g., FSCFNet-s, FSCFNet-m, and FSCFNet-l) also deliver competitive and enhanced performance, highlighting the scalability and robustness of the proposed modules under different computational budgets.

V. CONCLUSION

In this article, an efficient and lightweight detector called FSCFNet is designed, tailored for IRSTD. Specifically, three plug-and-play modules (FSCFNet, ACA, and MRCB) are proposed to enhance the feature representation. FSCFNet utilizes a multibranch structure based on the Haar WT, which effectively fuses both local fine-grained textures and global

frequency-aware spatial representations, while maintaining a low parameter count. ACA focuses on enhancing the central region of local small targets by generating asymmetric feature maps through convolutions with horizontal, vertical, and square kernels. MRCB captures long-range contextual dependencies by employing multiscale dilated convolutions, thereby enabling better focus on small object regions and improving the understanding of global feature relationships. In addition, WDL is adopted as the loss function to alleviate the limitations caused by the sparsity and discreteness of local infrared small target patterns. Extensive comparisons demonstrate that FSCFNet generally outperforms eight representative convolution modules, and ACA surpasses conventional attention mechanisms. Moreover, experimental results indicate that FSCFNet outperforms 18 competitive detection frameworks across three recognized datasets, confirming its high generalization, robustness, and efficiency. Comprehensive ablation and generalization studies further validate the effectiveness of the proposed modules. Future work could explore multimodal fusion strategies and advanced background suppression techniques to address more challenging remote sensing scenarios.

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